Solving Oblique Triangles with the Law of Cosines
Answer Key

Precalculus

Quick Answers

1. 7.24
2. 95.74
3. 9.24
4. 45.21
5. 17.29, 50.74
6. 55.86
7. 21.79, $\frac{13}{14}$
8. 36.28
9. 121.03
10. 59.49
11. 24.05, 50.58
12. 37.36

Curriculum

Level: Precalculus
HSG-SRT.D.10 – *problems 1, 2, 3, 4, 5, 7, 8, 10, 11*
HSG-SRT.D.11 – *problems 6, 9, 12*
Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

1. *If they got 14.41: wrote a plus sign in front of the 2ab cosine term instead of a minus*
4. *If they got 32.16: put the wrong side opposite the angle, using the c squared term as the subtracted one*
6. *If they got 95: added the two distances instead of using the law of cosines*

1. SAS: $a = 7$, $b = 9$, $C = 52°$; find c. Two sides and the angle *between* them is the SAS case, so the law of cosines applies directly with the unknown side opposite the known angle.

$$c^2 = 7^2 + 9^2 - 2(7)(9)\cos 52° = 49 + 81 - 126(0.615661)$$
$$c^2 = 130 - 77.5733 = 52.4267$$

Take the positive square root. $\boxed{c \approx 7.24}$
The answer is shorter than both given sides, which is right for an angle under 60° squeezed between them.

2. SSS: $a = 5$, $b = 7$, $c = 9$; find C. Three sides is SSS, so rearrange the law of cosines for the angle opposite the side you name — here c, the longest side.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{25 + 49 - 81}{2(5)(7)} = \frac{-7}{70} = -0.1$$

A negative cosine means an obtuse angle, and the inverse cosine returns it directly (no ambiguity, unlike the law of sines). $\boxed{C \approx 95.74^\circ}$

3. SAS: $b = 12$, $c = 15$, $A = 38^\circ$; find a. The angle sits between the two given sides, so the same formula applies with the letters rotated.

$$a^2 = 12^2 + 15^2 - 2(12)(15)\cos 38^\circ = 144 + 225 - 360(0.788011)$$
$$a^2 = 369 - 283.684 = 85.316$$

$\boxed{a \approx 9.24}$

4. SSS: $a = 8$, $b = 11$, $c = 6$; find A. Solve for the angle opposite a, so a^2 is the term that gets subtracted:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{121 + 36 - 64}{2(11)(6)} = \frac{93}{132} = 0.704545$$

$\boxed{A \approx 45.21^\circ}$

Check against the figure: $a = 8$ is the middle side, so its angle should be the middle angle — and 45.21° sits between the angles opposite 6 and 11.

5. SAS with area: $a = 9$, $b = 12$, $C = 110^\circ$. (a) The included angle is obtuse, so its cosine is negative and the subtraction becomes an addition — the third side comes out longer than both givens, as an obtuse triangle requires.

$$c^2 = 81 + 144 - 2(9)(12)\cos 110^\circ = 225 - 216(-0.342020)$$
$$c^2 = 225 + 73.876 = 298.876$$

$\boxed{c \approx 17.29}$

(b) The area needs no new triangle solving: two sides and the angle between them give Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(9)(12)\sin 110^\circ = 54(0.939693)$. $\boxed{\text{Area} \approx 50.74}$

6. Two ships, 40 km and 55 km apart at 70°. The bearings 030° and 100° differ by 70°, and that difference is the angle at the port between the two courses. The two courses and the line joining the ships form an SAS triangle.

$$c^2 = 40^2 + 55^2 - 2(40)(55)\cos 70^\circ = 1600 + 3025 - 4400(0.342020)$$
$$c^2 = 4625 - 1504.89 = 3120.11$$

$\boxed{c \approx 55.86 \text{ km}}$

Note that the ships end up almost exactly as far from each other as the longer voyage was long — a useful sense-check that the angle was read as the *difference* of the bearings.

7. SSS: $a = 6$, $b = 10$, $c = 14$. (a) Rearranging for the angle opposite a and keeping the fraction exact:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{100 + 196 - 36}{2(10)(14)} = \frac{260}{280}$$

Dividing numerator and denominator by 20 reduces it. $\boxed{\cos A = \frac{13}{14}}$

(b) Taking the inverse cosine of that exact value: $\boxed{A \approx 21.79^\circ}$
A cosine close to 1 means a small angle, which matches $a = 6$ being much the shortest side.

8. SAS with an obtuse included angle: $b = 23$, $c = 18$, $A = 124^\circ$.

$$a^2 = 23^2 + 18^2 - 2(23)(18)\cos 124^\circ = 529 + 324 - 828(-0.559193)$$
$$a^2 = 853 + 463.01 = 1316.01$$

$\boxed{a \approx 36.28}$
The sign is the whole story here: $\cos 124^\circ$ is negative, so the formula adds. A student who drops the sign gets about 19.75 and a triangle that cannot exist with a 124° angle.

9. Pilot: 120 mi, turn, 85 mi, interior angle 70°. The two legs of the flight and the direct line back to the airfield form a triangle whose known angle lies between the two known sides.

$$c^2 = 120^2 + 85^2 - 2(120)(85)\cos 70^\circ = 14400 + 7225 - 20400(0.342020)$$
$$c^2 = 21625 - 6977.21 = 14647.79$$

$\boxed{c \approx 121.03 \text{ mi}}$

10. SSS: $a = 13$, $b = 14$, $c = 15$; find B. B is opposite b, so b^2 is the subtracted term:

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{169 + 225 - 196}{2(13)(15)} = \frac{198}{390} = 0.507692$$

$\boxed{B \approx 59.49^\circ}$
The three sides are nearly equal, so all three angles should land near 60° — and this one does.

11. SAS then an angle: $a = 25$, $b = 32$, $C = 48°$. **(a)** Law of cosines for the side opposite the known angle:

$$\begin{aligned} c^2 &= 625 + 1024 - 2(25)(32)\cos 48° = 1649 - 1600(0.669131) \\ c^2 &= 1649 - 1070.61 = 578.39 \end{aligned}$$

$\boxed{c \approx 24.05}$

(b) With all three sides known, use the law of cosines again rather than the law of sines — the inverse cosine cannot land on the wrong angle, while the inverse sine of an obtuse angle returns its acute partner.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{1024 + 578.39 - 625}{2(32)(24.05)} = 0.635533$$

$\boxed{A \approx 50.58°}$

Accept the law of sines for (b) as well, provided the student checks which of the two candidate angles is correct; here A is acute because it is not opposite the longest side.

12. Survey crew: 18 **km, turn,** 26 **km, interior angle** 115°. **(a)** SAS again, with an obtuse included angle, so the cosine term adds:

$$\begin{aligned} c^2 &= 18^2 + 26^2 - 2(18)(26)\cos 115° = 324 + 676 - 936(-0.422618) \\ c^2 &= 1000 + 395.57 = 1395.57 \end{aligned}$$

$\boxed{c \approx 37.36 \text{ km}}$

(b) Reasoning answer — review by hand. A model response: the law of sines pairs each side with the angle opposite it, so using it requires a side-and-opposite-angle pair to be known. Here the only known angle, 115°, sits between the two known sides and its opposite side is exactly what is being asked for, so every ratio the law of sines could write contains two unknowns. The law of cosines needs no such pair — it relates all three sides to one angle — so it supplies the missing third side. Once that side is known there is a complete side-angle pair, and the law of sines becomes the quicker tool for the two remaining angles.

Credit any answer that identifies the missing side-opposite-angle pair as the obstacle and the third side as what the law of cosines supplies.